mobile robots. In the following, the geometrical features of the two classes are presented.

1.2.1 Robot Manipulators

The mechanical structure of a *robot manipulator* consists of a sequence of rigid bodies (*links*) interconnected by means of articulations (*joints*); a manipulator is characterized by an *arm* that ensures mobility, a *wrist* that confers dexterity, and an *end-effector* that performs the task required of the robot.

The fundamental structure of a manipulator is the serial or *open kinematic chain*. From a topological viewpoint, a kinematic chain is termed open when there is only one sequence of links connecting the two ends of the chain. Alternatively, a manipulator contains a *closed kinematic chain* when a sequence of links forms a loop.

A manipulator's mobility is ensured by the presence of joints. The articulation between two consecutive links can be realized by means of either a *prismatic* or a *revolute* joint. In an open kinematic chain, each prismatic or revolute joint provides the structure with a single degree of freedom (DOF). A prismatic joint creates a relative translational motion between the two links, whereas a revolute joint creates a relative rotational motion between the two links. Revolute joints are usually preferred to prismatic joints in view of their compactness and reliability. On the other hand, in a closed kinematic chain, the number of DOFs is less than the number of joints in view of the constraints imposed by the loop.

The *degrees of freedom* should be properly distributed along the mechanical structure in order to have a sufficient number to execute a given task. In the most general case of a task consisting of arbitrarily positioning and orienting an object in three-dimensional (3D) space, *six* DOFs are required, three for positioning a point on the object and three for orienting the object with respect to a reference coordinate frame. If more DOFs than task variables are available, the manipulator is said to be *redundant* from a kinematic viewpoint.

The *workspace* represents that portion of the environment the manipulator's end-effector can access. Its shape and volume depend on the manipulator structure as well as on the presence of mechanical joint limits.

The task required of the arm is to position the wrist which then is required to orient the end-effector. The type and sequence of the arm's DOFs, starting from the base joint, allows a classification of manipulators as *Cartesian*, *cylindrical*, *spherical*, *SCARA*, and *anthropomorphic*.

Cartesian geometry is realized by three prismatic joints whose axes typically are mutually orthogonal (Fig. 1.2). In view of the simple geometry, each DOF corresponds to a Cartesian space variable and thus it is natural to perform straight motions in space. The Cartesian structure offers very good mechanical stiffness. Wrist positioning accuracy is constant everywhere in the workspace. This is the volume enclosed by a rectangular parallel-piped

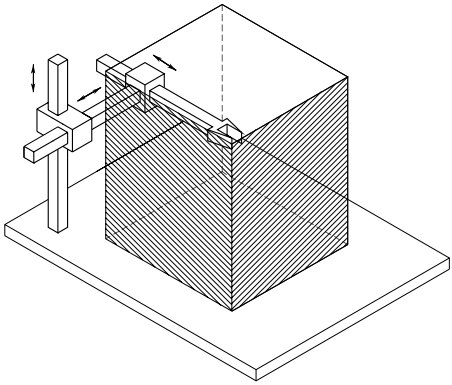


Fig. 1.2. Cartesian manipulator and its workspace

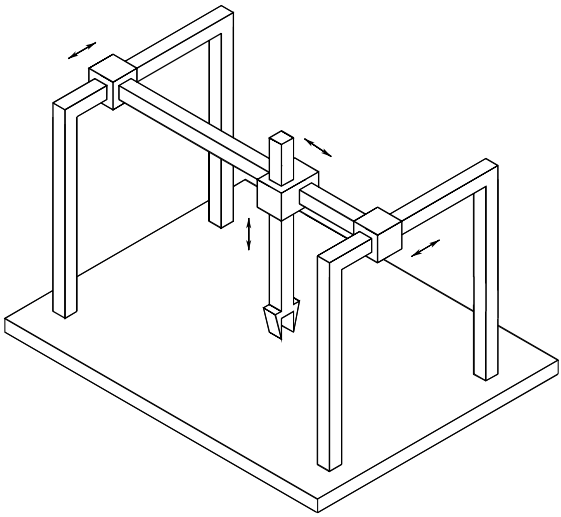


Fig. 1.3. Gantry manipulator

(Fig. 1.2). As opposed to high accuracy, the structure has low dexterity since all the joints are prismatic. The direction of approach in order to manipulate an object is from the side. On the other hand, if it is desired to approach an object from the top, the Cartesian manipulator can be realized by a *gantry* structure as illustrated in Fig. 1.3. Such a structure makes available a workspace with a large volume and enables the manipulation of objects of large dimensions and heavy weight. Cartesian manipulators are employed for material handling and assembly. The motors actuating the joints of a Cartesian manipulator are typically electric and occasionally pneumatic.

Cylindrical geometry differs from Cartesian in that the first prismatic joint is replaced with a revolute joint (Fig. 1.4). If the task is described in cylindri-

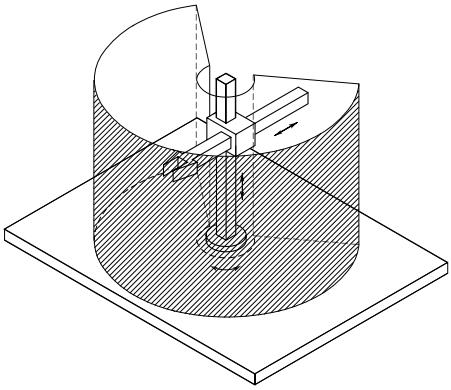


Fig. 1.4. Cylindrical manipulator and its workspace

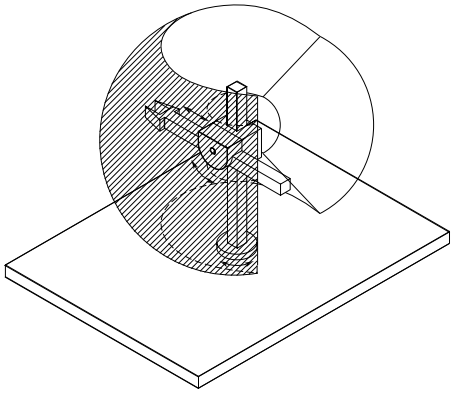


Fig. 1.5. Spherical manipulator and its workspace

cal coordinates, in this case each DOF also corresponds to a Cartesian space variable. The cylindrical structure offers good mechanical stiffness. Wrist positioning accuracy decreases as the horizontal stroke increases. The workspace is a portion of a hollow cylinder (Fig. 1.4). The horizontal prismatic joint makes the wrist of a cylindrical manipulator suitable to access horizontal cavities. Cylindrical manipulators are mainly employed for carrying objects even of large dimensions; in such a case the use of hydraulic motors is to be preferred to that of electric motors.

Spherical geometry differs from cylindrical in that the second prismatic joint is replaced with a revolute joint (Fig. 1.5). Each DOF corresponds to a Cartesian space variable provided that the task is described in spherical coordinates. Mechanical stiffness is lower than the above two geometries and mechanical construction is more complex. Wrist positioning accuracy decreases as the radial stroke increases. The workspace is a portion of a hollow sphere (Fig. 1.5); it can also include the supporting base of the manipulator and thus

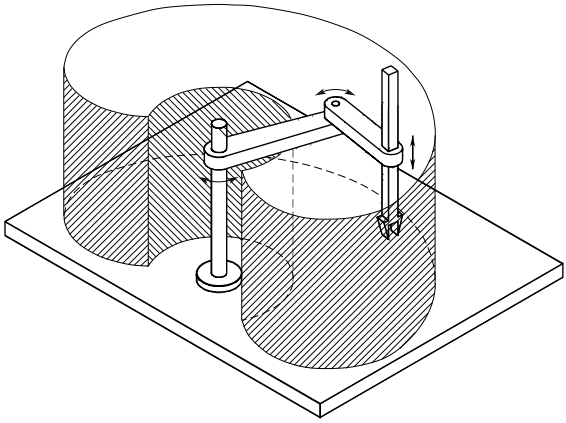


Fig. 1.6. SCARA manipulator and its workspace

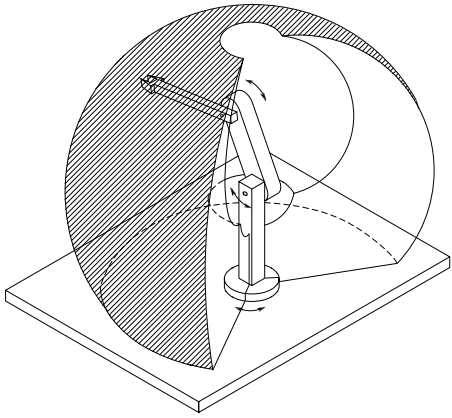


Fig. 1.7. Anthropomorphic manipulator and its workspace

it can allow manipulation of objects on the floor. Spherical manipulators are mainly employed for machining. Electric motors are typically used to actuate the joints.

A special geometry is *SCARA* geometry that can be realized by disposing two revolute joints and one prismatic joint in such a way that all the axes of motion are parallel (Fig. 1.6). The acronym SCARA stands for *Selective Compliance Assembly Robot Arm* and characterizes the mechanical features of a structure offering high stiffness to vertical loads and compliance to horizontal loads. As such, the SCARA structure is well-suited to vertical assembly tasks. The correspondence between the DOFs and Cartesian space variables is maintained only for the vertical component of a task described in Cartesian coordinates. Wrist positioning accuracy decreases as the distance of the wrist from the first joint axis increases. The typical workspace is illustrated

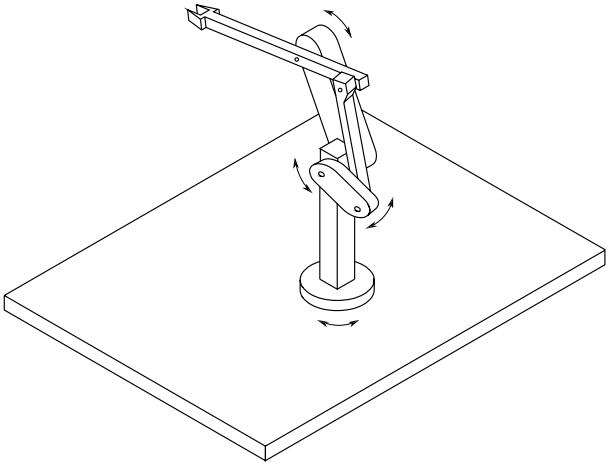


Fig. 1.8. Manipulator with parallelogram

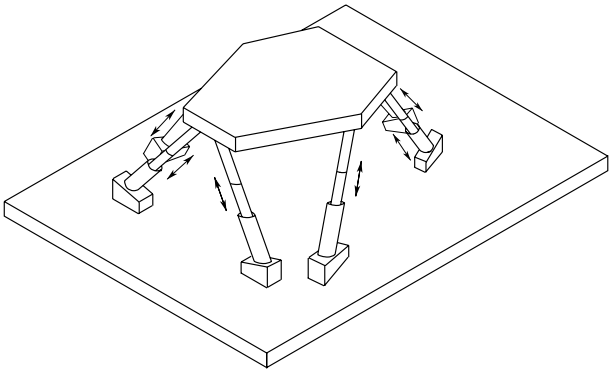


Fig. 1.9. Parallel manipulator

in Fig. 1.6. The SCARA manipulator is suitable for manipulation of small objects; joints are actuated by electric motors.

Anthropomorphic geometry is realized by three revolute joints; the revolute axis of the first joint is orthogonal to the axes of the other two which are parallel (Fig. 1.7). By virtue of its similarity with the human arm, the second joint is called the shoulder joint and the third joint the elbow joint since it connects the "arm" with the "forearm." The anthropomorphic structure is the most dexterous one, since all the joints are revolute. On the other hand, the correspondence between the DOFs and the Cartesian space variables is lost, and wrist positioning accuracy varies inside the workspace. This is approximately a portion of a sphere (Fig. 1.7) and its volume is large compared to manipulator encumbrance. Joints are typically actuated by electric motors. The range of industrial applications of anthropomorphic manipulators is wide.

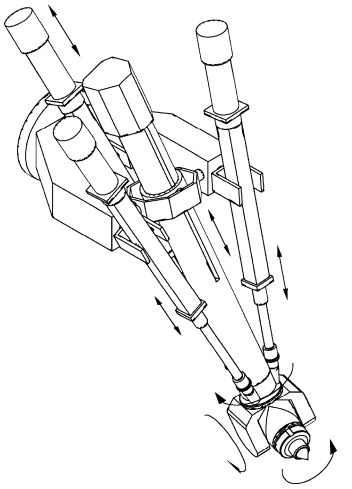


Fig. 1.10. Hybrid parallel-serial manipulator

According to the latest report by the *International Federation of Robotics* (IFR), up to 2005, 59% of installed robot manipulators worldwide has anthropomorphic geometry, 20% has Cartesian geometry, 12% has cylindrical geometry, and 8% has SCARA geometry.

All the previous manipulators have an open kinematic chain. Whenever larger payloads are required, the mechanical structure will have higher stiffness to guarantee comparable positioning accuracy. In such a case, resorting to a closed kinematic chain is advised. For instance, for an anthropomorphic structure, parallelogram geometry between the shoulder and elbow joints can be adopted, so as to create a closed kinematic chain (Fig. 1.8).

An interesting closed-chain geometry is *parallel* geometry (Fig. 1.9) which has multiple kinematic chains connecting the base to the end-effector. The fundamental advantage is seen in the high structural stiffness, with respect to open-chain manipulators, and thus the possibility to achieve high operational speeds; the drawback is that of having a reduced workspace.

The geometry illustrated in Fig. 1.10 is of hybrid type, since it consists of a parallel arm and a serial kinematic chain. This structure is suitable for the execution of manipulation tasks requiring large values of force along the vertical direction.

The manipulator structures presented above are required to position the wrist which is then required to orient the manipulator's end-effector. If arbitrary orientation in 3D space is desired, the wrist must possess at least three DOFs provided by revolute joints. Since the wrist constitutes the terminal part of the manipulator, it has to be compact; this often complicates its mechanical design. Without entering into construction details, the realization endowing the wrist with the highest dexterity is one where the three revolute

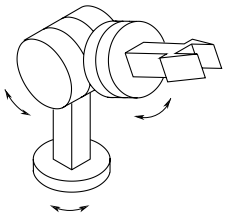


Fig. 1.11. Spherical wrist

axes intersect at a single point. In such a case, the wrist is called a *spherical wrist*, as represented in Fig. 1.11. The key feature of a spherical wrist is the decoupling between position and orientation of the end-effector; the arm is entrusted with the task of positioning the above point of intersection, whereas the wrist determines the end-effector orientation. Those realizations where the wrist is not spherical are simpler from a mechanical viewpoint, but position and orientation are coupled, and this complicates the coordination between the motion of the arm and that of the wrist to perform a given task.

The *end-effector* is specified according to the task the robot should execute. For material handling tasks, the end-effector consists of a gripper of proper shape and dimensions determined by the object to be grasped (Fig. 1.11). For machining and assembly tasks, the end-effector is a tool or a specialized device, e.g., a welding torch, a spray gun, a mill, a drill, or a screwdriver.

The versatility and flexibility of a robot manipulator should not induce the conviction that all mechanical structures are equivalent for the execution of a given task. The choice of a robot is indeed conditioned by the application which sets constraints on the workspace dimensions and shape, the maximum payload, positioning accuracy, and dynamic performance of the manipulator.

1.2.2 Mobile Robots

The main feature of *mobile robots* is the presence of a mobile base which allows the robot to move freely in the environment. Unlike manipulators, such robots are mostly used in service applications, where extensive, autonomous motion capabilities are required. From a mechanical viewpoint, a mobile robot consists of one or more rigid bodies equipped with a *locomotion* system. This description includes the following two main classes of mobile robots:[1]

- *Wheeled* mobile robots typically consist of a rigid body (*base* or *chassis*) and a system of wheels which provide motion with respect to the ground.

[1] Other types of mechanical locomotion systems are not considered here. Among these, it is worth mentioning *tracked locomotion*, very effective on uneven terrain, and *undulatory locomotion*, inspired by snake gaits, which can be achieved without specific devices. There also exist types of locomotion that are not constrained to the ground, such as flying and navigation.